Kushagra Shrivastava

Email LinkedIn Website

Skills and Interests

Observational Astronomy, Asteroseismology, Cosmology, Computational Modeling, Data Analysis, Science Communication, Competition Organisation

Education

National University of Singapore 2023 — Present
Bachelor of Science in Physics (Hons.)
Specialisation in Astrophysics
Special Program in Science
NUS College
Year 3 Sem 2 CGPA: 4.77/5.00

University of Copenhagen (Københavns Universitet) Spring 2026
Semester-long Exchange Student
Relevant Coursework: Galactic Dynamics and Astrobiology
GPA: 12/12 (Danish System)

Research Experience

Period-spacing analysis methods for Gamma Doradus stars 2026—Present
NUS AI, Data Science, and Informatics in Astronomy (NADIA), Singapore
Advisor: A. Prof. Marc Hon

Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms 2025
Satellite Technology and Research Center, Singapore
Advisor: A. Prof. Abhishek Rai

Fano Resonant Metasurfaces for Virus Testing Application 2021—2022
*A*STAR Institute of High-Performance Computing, Singapore*
Advisor: Dr. Hong-Son Chu

Growth Analysis of Sunspot in AR12741 using High-Resolution Solar Imaging
National Junior College, Singapore 2021
Advisors: Mr. Alfred Tan and Ms. Ning Hwee Tiang

Closing-up onto our sun: Solar Imaging 2019—2020
National Junior College, Singapore
Advisors: Mr. Alfred Tan and Ms. Ning Hwee Tiang
DOI: https://doi.org/10.1051/epjconf/202024007011

Research Schools

GraSPA (Graduate School for Particle and Astroparticle Physics) 2026
Laboratoire d'Annecy de Physique des Particules, Annecy, France

Teaching and Mentoring

CS1010S Introduction to Programming Methodology 2025 — Present
Teaching Assistant
National University of Singapore

SP2273 Working on Interdisciplinary Science, Pythonically 2025 — Present
Senior Mentor and Teaching Assistant
National University of Singapore

SP3176 The Universe 2025 — Present
Senior Mentor and Teaching Assistant
National University of Singapore

SP3275 Science for a Sustainable Earth 2025 — Present
Senior Mentor and Teaching Assistant
National University of Singapore

Professional Memberships

Royal Astronomical Society 2023—Present
Elected Fellow (FRAS)

European Astronomical Society 2023—Present
Affiliated Member

Leadership Positions

Cosmic Cup 2023—Present
Founder, Administrator, Question Master
Long-Type Questions: Explosion and Expansion — Hubble Parameter calculation through Type Ia Supernovae, Vedic Knowledge — Practical Astronomy through Astrology's lens
Website: www.cosmiccup.org

National University of Singapore Astronomical Society
President, 27th Executive Committee 2024—2025
Director of Academia, 26th Executive Committee 2023—2024

AstroChallenge
Finals In-Charge 2025
Question Master 2024—2025
Long-Type Questions: MOND-ifynig Newton's Gravitation (2024), The Sound of Space — Baryonic Acoustic Oscillations (2025)
Website: www.astrochallenge.org

CodeDay
Assistant Regional Manager, Singapore 2023
Volunteer, Lucknow, India 2024—2025
Volunteer, Kolkata, India 2024
Website: www.codeday.org

Scholarships and Awards

NUS Science and Technology Merit Scholarship 2023—2027
Awarded by National University of Singapore
Full-ride scholarship (with other benefits) for undergraduate studies.

Rhodes Scholarship Finalist 2026
Awarded by the Rhodes Trust Singapore Constituency
Among the top 10 applicants from a country-wide and inter-jurisdictional pool in Singapore.

A*STAR Youth Scholarship 2019—2022
*Awarded by A*STAR and Ministry of Education, Singapore*
Full-ride scholarship (with other benefits) for Grade 9 to Grade 12 studies.

Best Research Presentation at SPS Congress 2025
Awarded by Special Program in Science
Awarded for best presentation on "Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms"

Finalist, International Natural Sciences Tournament 2024
Awarded by International Natural Science Tournament Committee

Second Runner-up, PLANCKS Singapore Preliminaries 2024
Awarded by Institute of Physics, Singapore, Student Chapter

Sembcorp Marine Prize 2023
Awarded by National University of Singapore
Awarded for being the "Best Student in Physics"

Best Research in Sustainable Material Category 2022
Awarded by National Junior College
Awarded for "Fano Resonant Metasurfaces for Virus Testing Application" research.

Gold Prize, International Geosciences Challenge 2022
Awarded by Asia Oceania Geosciences Society
Awarded for "Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging" research presentation.

Merit and Commendation Awards, NJC STaR Symposium 2021
Awarded by National Junior College
Awarded for "Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging" research.

Merit Award, NJC STaR Symposium 2020
Awarded by National Junior College
Awarded for "Closing-up onto our sun: Solar Imaging" research.

Conferences

NUS Special Program in Science Research Congress 2025
Presented *Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms.*

NJC STaR Symposium (Virtual) 2022
Presented *Fano Resonant Metasurfaces for Virus Testing Application.*

Singapore Science and Engineering Fair (Virtual) 2022
Presented *Fano Resonant Metasurfaces for Virus Testing Application.*

Thai Astronomical Conference (Virtual) 2022
Presented *Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.*

Singapore Science and Engineering Fair (Virtual) 2021
Presented *Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.*

Asia-Oceania Geosciences Society AGM (Virtual) 2021
Presented *Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.*

NJC STaR Symposium (Virtual) 2021
Presented *Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.*

NJC STaR Symposium 2020
Presented *Closing-up onto our sun: Solar Imaging.*

South-East Asian Astronomy Networking Meeting 2019
Presented *Closing-up onto our sun: Solar Imaging.*